

AI-Enhanced Educational Management System Feasibility Study

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Section 1.0: Statement Of Work

1.1 An Introduction to the AI-Enhanced Educational Management System

Amid global digital transformation in education, academic institutions still rely on fragmented, manual systems for student records, scheduling, grading, and faculty management, causing inefficiency and data silos. Traditional platforms lack real-time tracking, multi-dimensional evaluation, and academic early warning. Advances in AI, especially large language models and retrieval-augmented generation, now enable intelligent upgrades beyond simple automation. This project will design and deliver a fully integrated, AI-enhanced educational management system to unify core workflows, eliminate silos, and embed AI throughout academic operations. It will serve administrators, teachers, students, and guardians, reducing manual burdens, improving efficiency and teaching quality, and supporting personalized student development.

1.2 Project Scope Definition

The AI-EMS project scope defines all in-scope deliverables, functional modules, users, technical execution, lifecycle activities, and explicit exclusions to prevent scope creep and ensure on-time, on-budget delivery meeting quality requirements.

In-Scope Elements

Functional Modules (6 core modules):

1. **Intelligent Teaching Assistance:** 24/7 LLM-powered Q&A with multi-turn dialogue and policy/document linking.
2. **Integrated Attendance Management:** Real-time schedule sync, anomaly alerts, location verification, online leave requests.
3. **AI Course Evaluation:** Anonymous multi-dimensional ratings and open-ended feedback integrated into the knowledge base.
4. **Academic Overview & Intelligent Planning:** Student dashboards (credits, GPA trends, personalized recommendations); teacher analytics for at-risk identification.
5. **Basic Information Management:** Role-based management of student/teacher/course data with review workflows.
6. **System Integration & Data Ecosystem:** SSO, standardized APIs for third-party integration and future AI expansion.

User Scope: Academic administrators (statistics, approvals, monitoring), teaching staff (course management, attendance, analytics), students (schedule, progress, selection, leave, evaluation).

Technical Scope: Cross-platform web/mobile app using Uniapp + Node.js, DeepSeek API (LLM), Pinecone (vector search), LangChain (agentic workflows), Opendb (JSON), microservices-ready layered architecture, Unicloud deployment, GitHub version control, Docker for consistency.

Lifecycle Execution Scope: Scrum-based activities including requirements/PRD, UI/UX design, iterative development, AI/RAG integration, testing, documentation (feasibility study, user manual, final report), and PoC delivery/presentation.

Deliverables: Complete project documentation, deployable AI-EMS PoC software, group workload profile and assessment materials.

Explicit Out-of-Scope Items (The following are strictly excluded from development, delivery, and maintenance):

Financial Systems: No tuition collection, salary processing, invoicing, or any financial transaction handling.

Live Teaching Infrastructure: No built-in video conferencing, lecture streaming, or real-time online classroom integration (e.g., Zoom, Teams).

Hardware Procurement/Integration: No development, purchase, or maintenance of physical devices (biometric scanners, card readers, attendance

terminals).

Full Academic Content Generation: AI responses limited to verified knowledge base; no auto-generation of full textbooks, exam papers, or official credentials.

Non-Academic Campus Management: Excludes dormitory, logistics, campus security, and student activity administration.

Post-Delivery Maintenance: Long-term system upkeep, version iteration, and daily operations are not covered beyond PoC handover.

Compliance Certification: No formal accreditation or legal compliance auditing (e.g., FERPA, GDPR) is included in scope.

Business Planning: No development of commercialization strategy, go-to-market roadmap, business model design, or investor pitch materials.

1.3 Key Functional and Performance Requirements

To ensure the AI-Enhanced Educational Management System (AI-EMS) effectively addresses the identified administrative bottlenecks and delivers seamless user experience, the system must adhere to the following specific functional and performance baselines during its development and deployment.

Functional Requirement

The system logic must be strictly bound by the following functional rules across its core modules:

1. **Intelligent Teaching Assistance (AI Q&A):** The LLM-driven assistant must maintain conversational context for a minimum of five continuous interactions. It must be capable of extracting and synthesizing information exclusively from the verified institutional knowledge base. In scenarios where a user's query falls outside the knowledge base, the system must trigger a default fallback protocol, clearly stating the limitation rather than generating unverified responses, thereby mitigating the risk of AI hallucination¹.
2. **Integrated Attendance and Workflows:** The location-based attendance check-in must enforce a geofencing tolerance (e.g., within a 50-meter radius of the designated classroom coordinates). The leave application workflow must be fully automated: upon administrator or counselor approval, the system must immediately and automatically overwrite the student's default "absent" status to "on leave" in the respective course attendance logs.
3. **AI-Powered Course Evaluation:** The evaluation system must enforce strict anonymity at the database level, irreversibly decoupling student identity tokens from submitted feedback. Upon submission, textual evaluation data must be vectorized and updated into the Pinecone database to ensure the AI Q&A module accesses the most current peer-review data.
4. **Academic Planning & Early Warning:** The personalized course recommendation algorithm must compute suggestions based on a mandatory triangulation of three data points: the student's completed course grades, self-selected interest tags, and historical course evaluation scores. For teaching staff, the dashboard must automatically flag "at-risk" students when specific quantitative thresholds are breached, effectively applying learning analytics to improve early intervention and student retention rates².
5. **Role-Based Data Management:** The system must implement stringent Role-Based Access Control (RBAC). Core identifiers (e.g., Student ID, Major) must remain immutable to end-users. Modifications to secondary personal information initiated by students must be placed in a pending state and require explicit administrative approval before database commits occur.

Performance and Architectural Requirements

The system infrastructure must be engineered to guarantee stability, responsiveness, and seamless delivery:

1. **Capacity and Concurrency:** The infrastructure must robustly support the Proof of Concept (PoC) initial scale of 2,200 active users. Furthermore, the architecture must possess the elasticity to absorb a minimum 35% instantaneous surge in concurrent traffic without experiencing service degradation or timeouts.
2. **Latency Thresholds:** Standard graphical interface interactions and CRUD (Create, Read, Update, Delete) operations must execute and render within 2.0 seconds under normal load, aligning with established human-computer interaction usability thresholds (Nielsen, 1993)³. For queries processed by the DeepSeek API, the system must achieve a Time-to-First-Token (TTFT) response of no more than 4.0 seconds to maintain conversational fluidity.
3. **Architectural Standards:** The application must strictly follow a decoupled, microservices-ready architecture. The frontend must be developed using Uniapp to ensure perfect parity between web and mobile interfaces. All deployment environments must be containerized using Docker to guarantee absolute consistency across development, testing, and production phases.

1.4 Acceptance Test Standards

The ultimate validation of the AI-EMS PoC will be determined by its ability to pass a comprehensive suite of acceptance criteria. The project will be deemed successful and ready for closure only when the following conditions are demonstrably met.

Role-Based User Acceptance Testing (UAT)

The software must successfully execute end-to-end operational loops for each specific user group without critical errors:

1. **Student Scenario:** A test student account must be able to log in, query the AI assistant regarding a specific graduation policy, navigate to the Academic Dashboard to view AI-recommended courses based on that policy, and successfully submit an anonymous evaluation for a previously completed course.
2. **Teacher Scenario:** A test instructor account must be able to access their assigned class, view a dynamically generated attendance report, and visually identify the system's automated red-flagging of students who have breached the predefined absence threshold.
3. **Administrator Scenario:** A test admin account must receive an automated notification for a student's profile update request, successfully execute an "Approve" action, and verify the data change is reflected globally across the system.

Performance and Stress Validation

The system must prove its resilience under simulated pressure. Using load-testing automation tools, the system must be subjected to simulated peak-week traffic volumes. The acceptance standard requires the system to maintain continuous uptime for a predefined test window with an API failure rate of less than 0.1%, while keeping cloud server CPU utilization within safe operational limits.

Architecture and Security Verification

Codebase and infrastructure audits must confirm the utilization of the proposed technology stack. Verification will ensure that the AI Agentic workflows

are genuinely routed through LangChain and Pinecone vector retrieval, rather than relying on legacy keyword database searches. Additionally, security audits must confirm that all user authentication credentials are cryptographically hashed and never stored in plain text.

Scope Boundary Verification

A critical negative-testing audit will be conducted to ensure the project has not succumbed to scope creep, which is an essential quality control practice in agile development frameworks⁴. The acceptance team will verify the absolute absence of explicitly excluded features. The project passes this criterion only if no financial payment gateways exist in the codebase, and no physical hardware SDKs are required to execute the attendance features.

Final Deliverables Check

Administrative acceptance requires the complete submission of all project artifacts. This includes the deployment-ready PoC software repository, this comprehensive Feasibility Study Report, the initial Project Proposal, the User Manual, and the definitive Group Workload Profile detailing the contributions of all CS Group 11 members.

Section 2.0: The Financials of the AI-Enhanced Educational Management System

2.1 Typical Costs of the System

The cost structure of the AI-Enhanced Educational Management System is divided into two core phases: the PoC prototype development phase (for this course project) and the full-scale deployment phase (for campus-wide rollout). All cost estimates are derived from peer-reviewed research on AI-enabled educational systems, which provide a standardized benchmark for campus-level system implementation^{5 6 7}.

For the course-level PoC prototype, the cost is highly optimized for small-scale validation, while the full-scale system follows the industry cost model for smart campus educational management systems, which covers development, infrastructure, AI service, and long-term operational components. The detailed cost breakdown is shown in Table 2.1.1.

Table 2.1.1: Typical Cost Breakdown of the AI-Enhanced Educational Management System

Cost Category	Cost Item	PoC Prototype Phase Cost (USD)	Full-Scale Deployment Phase Cost (USD)
One-Time Development Costs	Development Labor Cost	0 (Course group volunteer work)	8,500.00
	System Design & Testing Cost	0 (Course project internal work)	1,200.00
	Initial Domain & SSL Certificate Cost	15.00	15.00
Infrastructure Costs	Cloud Server Cost (Initial Setup)	50.00	350.00
	Database Initial Setup Cost	0 (Free open-source MySQL)	100.00
AI Service Costs	AI Model API Cost (Initial Training)	100.00	800.00

Cost Category	Cost Item	PoC Prototype Phase Cost (USD)	Full-Scale Deployment Phase Cost (USD)
	AI Agent Service Authorization Cost	0 (Free tier for small-scale)	1,200.00/Year
Tool & Third-Party Costs	Development Tool Subscription	0 (Free open-source tools)	300.00/Year
	Testing & Monitoring Tool Cost	0 (Free tier)	240.00/Year
Total One-Time Cost	-	165.00	11,065.00
Annual Recurring Cost	-	0 (PoC short-term)	1,740.00

2.2 Expected Pageviews of the System

The expected pageviews of the system are estimated based on the user behavior data of campus educational management systems from peer-reviewed studies, combined with the user scale of our target campus (12,000 students, 1,000 teachers, 100 administrative staff)^{8 9 10}.

According to the log analysis of campus information systems, the user behavior of educational management platforms presents a stable pattern: students have an average of 10.2 daily pageviews, teachers have an average of 18.5 daily pageviews, and administrative staff have an average of 22.3 daily pageviews⁸. Meanwhile, the system will have obvious peak traffic during the course selection week and the final exam week, when the pageviews will increase by about 35% compared to the normal week, which is consistent with the user behavior pattern observed in existing campus LMS systems¹⁰.

For the PoC prototype phase, we will conduct a small-scale pilot with 2,000 students and 200 teachers, the expected pageviews are shown in Table 2.2.1. After the full-scale deployment, the system will cover all campus users, and the expected pageviews will reach a stable level.

Table 2.2.1: Expected Pageviews of the AI-Enhanced Educational Management System

User Group	User Scale (PoC Phase)	Average Daily Pageviews Per User	Daily Total Pageviews (PoC Phase)	User Scale (Full-Scale Phase)	Daily Total Pageviews (Full-Scale Phase)
Students	2,000	10.2	20,400	12,000	122,400
Teachers	200	18.5	3,700	1,000	18,500
Administrative Staff	0	22.3	0	100	2,230
Total (Normal Week)	2,200	-	24,100	13,100	143,130
Total (Peak Week: Course Selection/Exam)	2,200	-	32,535	13,100	193,225.5

The expected pageviews show that the system can fully meet the traffic demand of the campus, and the cloud infrastructure we planned can easily handle the peak traffic during the course selection week, which is validated by the system scalability design we proposed in the previous system design section.

2.3 Operating costs of system

The operating costs of the AI-Enhanced Educational Management System (AI-EMS) refer to the recurring expenses required to maintain stable, secure, and scalable system operation after the initial deployment of the PoC prototype or full-scale rollout. These costs are estimated based on the expected pageviews (Section 2.2), the technical architecture defined in the Project Scope (Section 1.2), and the performance requirements (Section 1.3), including DeepSeek API usage for LLM inference, Pinecone vector database maintenance, Uniclass hosting, and basic operational support. All estimates follow industry benchmarks for cloud-based educational AI systems and are divided into PoC prototype phase (small-scale pilot with 2,200 users) and full-scale deployment phase (13,100 users). The detailed breakdown is presented in Table 2.3.1.

Table 2.3.1: Operating Costs of the AI-Enhanced Educational Management System

Cost Category	Cost Item	PoC Prototype Phase (Monthly USD)	Full-Scale Deployment Phase (Monthly USD)	Full-Scale Deployment Phase (Annual USD)	Notes / Basis
Cloud Infrastructure	Uniclass Hosting & Bandwidth	8.00	45.00	540.00	Based on 24,100–32,535 daily pageviews (PoC) and 143,130–193,226 (full-scale); includes containerized Docker instances and auto-scaling.
AI Service Costs	DeepSeek API Inference	12.00	85.00	1,020.00	Estimated 20–25% of pageviews trigger LLM Q&A (≈4,800–8,100 calls/day PoC; 28,600–48,300 calls/day full-scale); average 0.002–0.003 USD per query.
Vector Database	Pinecone Storage & Queries	5.00	25.00	300.00	Vector embeddings of institutional knowledge base + course evaluations; scales with user-generated feedback.
Data & Backup	Database Storage & Backup	3.00	12.00	144.00	OpenDB (JSON) + daily automated backups.
Monitoring & Maintenance	Logging, Monitoring & Basic Support	4.00	18.00	216.00	Includes Docker monitoring tools and minimal part-time DevOps support.
Total Operating Cost	–	32.00	185.00	2,220.00	Excludes one-time costs from Section 2.1.

As shown in the table, the PoC prototype incurs only USD 32 per month in operating costs, which can be fully covered by the course project budget. For full-scale campus deployment, the annual operating cost is approximately USD 2,220 (≈ USD 185/month). This figure is conservative and significantly lower than the average annual operating expenditure of commercial educational management platforms, primarily due to the use of cost-efficient APIs (DeepSeek), serverless-ready Uniclass deployment, and the microservices architecture that enables automatic scaling during peak periods (course selection/exam weeks).

These operating costs are expected to remain stable in the first year and can be further optimized through caching mechanisms for frequent Q&A queries and usage-based scaling of AI inference. Combined with the one-time costs in Section 2.1 and the clear benefits outlined in Section 5.0, the system demonstrates strong long-term economic feasibility.

Section 3.0: The Project Milestones

3.1 Project Stage Division

The AI-EMS project adopts a Scrum-based agile development framework (as defined in the Project Lifecycle Execution Scope in Section 1.2), which emphasizes iterative delivery, continuous integration, and close stakeholder feedback. The entire project is divided into five sequential yet overlapping stages to ensure systematic progress from planning to PoC delivery. Each stage includes clearly defined objectives, main deliverables, and responsible roles, aligning with the functional modules, technical scope, and acceptance criteria specified in Sections 1.2–1.4.

The project stages are as follows:

- 1.Planning and Requirements Gathering Stage** Focus: Finalize project scope, conduct user research with target stakeholders (students, teachers, administrators), and produce the Product Requirements Document (PRD) and this Feasibility Study Report. Key activities: Literature review, stakeholder interviews, functional/non-functional requirement definition. Deliverables: PRD, Feasibility Study Report, initial backlog. Duration: 2 weeks.
- 2.System Design Stage** Focus: Complete UI/UX prototyping, database schema design, system architecture (microservices-ready layered architecture), and AI workflow design (LangChain + Pinecone RAG). Key activities: Wireframing (Uniapp), API interface design, DeepSeek prompt engineering, security and RBAC design. Deliverables: UI/UX prototypes, architecture diagram, database schema, technical design document. Duration: 2 weeks.
- 3.Development and AI Integration Stage** Focus: Implement all six core functional modules iteratively using Scrum sprints (2-week cycles), integrate DeepSeek LLM, Pinecone vector search, and LangChain agentic workflows. Key activities: Frontend (Uniapp), backend (Node.js), AI/RAG pipeline development, SSO implementation, GitHub version control. Deliverables: Functional code repository, integrated AI Q&A and attendance modules, preliminary PoC version. Duration: 5 weeks (3 sprints).
- 4.Testing and Optimization Stage** Focus: Conduct comprehensive testing to meet the acceptance standards in Section 1.4. Key activities: Unit testing, integration testing, load testing (simulating 35% traffic surge), UAT with role-based scenarios, performance optimization (latency < 4s for AI responses). Deliverables: Test reports, bug-fixed PoC, optimized system. Duration: 3 weeks.
- 5.Deployment, Documentation and Closure Stage** Focus: Deploy the PoC to Uncloud, finalize all project documentation, and prepare for final presentation. Key activities: Docker containerization, deployment, user manual writing, workload profile compilation. Deliverables: Deployable PoC software, User Manual, Final Project Report, presentation materials. Duration: 2 weeks.

The stage division ensures that core AI-enhanced functions (especially the Intelligent Teaching Assistance and Academic Planning modules) are prioritized in early sprints, allowing incremental validation against the performance requirements in Section 1.3. This structure also provides sufficient buffer for risk mitigation (Section 4.0) and guarantees on-time delivery within the course timeline. Detailed key milestones and time feasibility assessment are provided in Sections 3.2 and 3.3 respectively

3.2 List of Key Milestones

- M1 | March 30, 2026 | Project Proposal Submission:** Complete requirements gathering, UI/UX prototyping, and architecture design; submit project proposal document.
- M2 | April 30, 2026 | Core Development Completion:** Build database schema; complete student/faculty profile management, basic attendance tracking, and UI framework.
- M3 | May 24, 2026 | AI Features Integration:** Integrate DeepSeek API; build RAG knowledge base; implement intelligent Q&A and course recommendation.

- M4 | June 1, 2026 | Final Report Submission:** Complete system testing and UI refinement; submit final report and user manual.
- M5 | June 8, 2026 | PoC Software Submission:** Submit deployable PoC system, complete user manual, and deployment documentation.
- M6 | June 15, 2026 | Final Presentation:** Conduct system demonstration to stakeholders; showcase core features.
- M7 | June 16, 2026 | Group Workload Profile Submission:** Submit signed Group Workload Profile with all 10 team members' confirmation.

3.3 Time Feasibility Assessment

The total project duration is from March 1, 2026, to June 16, 2026, spanning roughly 14 weeks (3.5 months), which strictly adheres to the course timeline. According to the stage division defined in Section 3.1, the five sequential phases are: Planning & Requirements (2 weeks), System Design (2 weeks), Development & AI Integration (5 weeks), Testing & Optimization (3 weeks), and Deployment, Documentation & Closure (2 weeks). Each phase is allocated a reasonable length, with the development phase occupying the highest proportion (5 weeks), ensuring iterative implementation of all six core modules. The testing phase reserves three weeks, sufficient for unit testing, integration testing, load testing (simulating 35% traffic surge), and user acceptance testing (UAT) under role-based scenarios, while also meeting the performance requirement of AI response latency below 4 seconds.

Key milestones (M1 to M7) are clearly defined, starting from the submission of the project proposal on March 30, 2026, to the final workload profile submission on June 16, 2026. The intervals between milestones are evenly spaced, enabling effective progress tracking and timely risk identification. The team adopts the Scrum agile framework, conducting bi-weekly sprints that allow parallel development of frontend (Uniapp), backend (Node.js), and AI components (DeepSeek API, Pinecone, LangChain). This approach efficiently manages the collaboration complexity of a 10-person team and ensures early delivery of high-priority features such as the intelligent Q&A assistant and academic planning dashboard.

To ensure on-time delivery despite uncertainties, the timeline incorporates multiple layers of risk buffer. First, each two-week Sprint reserves approximately 10% of capacity (i.e., one working day) as a “contingency slot” to handle unexpected issues such as API rate limits, debugging delays, or last-minute requirement clarifications. Second, the five-week development phase is deliberately designed to complete all core coding by the end of week four, leaving the fifth week exclusively for integration stabilization and preliminary internal testing. This prevents cascading delays into the formal testing phase. Third, the three-week testing phase includes one full week of buffer after UAT execution, allowing the team to fix critical bugs without compressing regression testing. Fourth, an eight-day gap exists between the deployment milestone (June 8) and the final presentation (June 16), which serves as a final buffer for performance tuning, documentation polishing, and presentation rehearsal.

Additionally, the team has identified specific high-risk items: DeepSeek API latency variability, potential schema changes in OpenDB, and cross-module integration conflicts (e.g., attendance + AI Q&A). For each, a pre-defined mitigation action is ready—such as prompt caching, fallback rule-based responses, and daily integration builds—further reducing the need for unplanned overtime. With these multi-layered buffers, even if one or two minor delays occur, the project remains on track to meet all milestones.

Since the project delivers a proof-of-concept (PoC) system without complex modules such as financial transactions, live video infrastructure, or hardware integration, and leverages mature technologies (Uniapp, Node.js, OpenDB, DeepSeek API), development efficiency is highly predictable. Therefore, completing this project within the given course timeline is highly feasible.

Section 4.0: Project's Risks

4.1 Risk Identification

- 1.The large model requires a large volume of training and operational data, and the collected data may involve intellectual property infringement or privacy violations.
- 2.Performance bottlenecks may occur during peak user periods, caused by scenarios such as multi-turn dialogues and generation of teaching policy documents.
- 3.Location verification functions may be bypassed by fake positioning tools, resulting in abnormal or invalid attendance check-in.
- 4.Teaching resources, internal documents, and sensitive data may be leaked due to incomplete development or insufficient security protection.
- 5.Improper permission management and weak system encryption may allow students to access teacher data or even the admin backend, causing damage to teaching management and resources (students may also act as hackers).
- 6.Database storage overflow or excessive data growth may lead to user data loss or system unavailability.
- 7.Third-party services may become unavailable due to server upgrades, service suspension, copyright updates, or policy changes, resulting in the failure of core system functions.
- 8.Frequent requirement changes may lead to project delays, rework, and unstable system delivery. Risk identification will be continuously updated throughout the project development lifecycle.

4.2 Risk Impact Assessment

- 1.**Data Infringement Risk:**Unauthorized data usage may lead to legal disputes, privacy compliance problems, financial penalties, and damage to the system's reputation.
- 2.**Peak Performance Bottleneck Risk:**Slow response, system lag, or timeouts will seriously reduce user experience and availability, affecting normal usage of core functions.
- 3.**Location Verification Bypass Risk:**Fake positioning will result in invalid attendance records, distorted data, and failure of teaching management rules.
- 4.**Teaching Resource Leakage Risk:**Internal documents and sensitive data leakage will cause intellectual property loss, security incidents, and compliance risks.
- 5.**Unauthorized Access Risk:**Illegal access to the backend or teacher data may lead to data tampering, system damage, and disorder in teaching management.

6.Database Storage Overflow Risk:Data loss, system crash, or service interruption will directly affect the stable operation of the entire system.

7.Third-Party Service Unavailability Risk:Core AI functions will be disabled, leading to system paralysis and failure of project delivery.

8.Requirement Change Risk:Frequent changes will cause rework, cost overruns, schedule delays, and unstable system delivery.

Risk impact assessment will be updated dynamically with project progress to ensure risk controllability.

4.3 Risk Identification

Table 4.3.1: Risk Countermeasures and Trigger Conditions

Risk Category	Risk Description	Trigger Conditions	Preventive Measures	Mitigation & Remedial
Technical Risks	AI hallucination & KB inaccuracy (Ji et al., 2023)	Hallucination >5%; ≥2 fallback failures; >3 wrong feedbacks/sprint	Constrain LLM to Pinecone KB; pre-validate content; weekly testing	Mitigate: Freeze scenarios, update KB; Remedy: Regression test, manual review
	Performance & concurrency failure	CRUD latency >2.0s; AI TTFT >4.0s; 35% surge degradation; API failure >0.1%	Microservices; caching; load testing; DB optimization	Mitigate: Auto-scaling; Remedy: Performance audit, 72h stress test
	Third-party API dependency	API availability <99%; quota >90%; interface incompatibility	SLA services; 30% quota buffer; offline fallback	Mitigate: Switch backup API; Remedy: Negotiate compensation
	Unauthorized access	Successful access; role breach; core ID modification	Strict RBAC; credential hashing; bi-weekly penetration testing	Mitigate: Freeze accounts; Remedy: Security audit, fix loopholes
Project Management Risks	Scope creep (Schwaber & Sutherland, 2020)	Out-of-scope development; changes affecting milestones/>10% workload	Strict change control; clear scope communication	Mitigate: Suspend out-of-scope work; Remedy: Reprioritize backlog
	Schedule delay	Sprint completion <90% (2x); milestone delay >3 days	Granular tasks; 2-week buffer; daily tracking	Mitigate: Reprioritize core tasks; Remedy: Adjust timeline
Cost & Resource Risks	Cost overrun	Overrun >10%; monthly cost >20% (2x); quota overrun	Budget tracking; cost caps; API optimization	Mitigate: Use emergency budget; Remedy: Cost audit
User & Business Adoption Risks	Usability & adoption defects	Task completion <95%; >3 usability feedbacks/role	User-centric design; iterative feedback	Mitigate: Fix critical defects; Remedy: UX audit, training

All the countermeasures we have formulated can be implemented within the project's time, resource and budget constraints(shown in the table4.3.1).As for those low-priority risks with minor impacts, we only keep a regular watch on them and record them, without devoting active resources to handle them.

4.4 Risk Monitoring Mechanism

To ensure effective risk control throughout the project lifecycle, we establish a Scrum-integrated monitoring mechanism with clear responsibilities, cycles, indicators, and processes. The Scrum Master oversees all monitoring activities, chairs risk reviews, and approves response plans; module owners monitor their respective module risks, report status, and execute mitigations; any team member must report newly identified risks within 24 hours; and the liaison collects user feedback to feed into the monitoring system. The monitoring cycle operates at three levels aligned with Scrum: daily updates during the 15-minute stand-up to capture sudden risks, phase-gate reviews before transitioning between stages to resolve high-priority risks, and sprint retrospective meetings to reassess risk status. Quantifiable indicators across five dimensions guide objective monitoring, as shown in Table 4.4.1.

Table 4.4.1: Quantifiable Risk Monitoring Indicators

Dimension	Core Indicators	Threshold	Data Source
Technical	Hallucination rate, latency, API failure	Hallucination >5%; latency >2.0s; API failure >0.1%	Test logs, UAT results
Project Management	Scope creep, sprint completion, delay	Out-of-scope work; completion <90%; delay >3 days	GitHub, backlog
Security	Unauthorized access, data encryption	Successful access; plaintext credentials	Security logs, DB records
Cost & Resource	Cost deviation, utilization, quota	Overrun >10%; CPU >80%; quota >90%	Billing reports
User Adoption	Task completion, usability feedback	Completion <95%; >3 feedbacks/role	UAT, stakeholder feedback

The Scrum Master consolidates and updates these indicators during each sprint review. We follow a standardized five-step closed-loop process: report risks via GitHub Issues within 24 hours, assess priority within two business days, assign owners for medium/high-priority risks, track progress in daily stand-ups while adjusting strategies as needed, and finally verify closure and archive resolved risks. For tooling, we use GitHub Issues with risk labels for tracking, real-time dashboards (Docker, Uniclass, API) with alerts, version-controlled document archiving, and regular knowledge updates to refine indicators and best practices. This integrated mechanism ensures risks are identified, managed, and resolved without disrupting project progress.

Section 5.0: The Benefits of the AI-Enhanced Educational Management System

5.1 Benefits Summary

The AI-EMS transforms traditional academic operations through AI integration and a unified digital platform. It automates routine tasks, breaks down data silos, and unlocks actionable insights from academic records. Students receive personalized guidance, teachers reclaim time for instruction, and administrators gain streamlined oversight. The system refocuses institutional energy on educational quality rather than manual coordination. Cross-platform design and single sign-on ensure intuitive access, while robust permission controls safeguard data privacy. Open APIs and scalable architecture enable future expansion and continued digital evolution.

5.2 Students Benefits

- 1. 24/7 smart portal:** Cross-platform unified entrance with AI-powered assistant using LLM to answer real-time questions on policies,

registration, schedules, exams, and graduation. Supports multi-turn dialogue and single sign-on, eliminating fragmented logins and slow information retrieval.

2. Academic visualization: Personalized dashboard with progress bars and radar charts dynamically tracking credit completion for general and specialized courses. Integrates GPA trends and percentile rankings for immediate, intuitive academic oversight.

3. Scientific planning: System recommends elective courses and potential paths (e.g., data analysis) based on performance history, completed courses, and interest tags, while clarifying relevance, difficulty, and prerequisites to reduce blind selection.

4. Objective reference: Anonymous, multi-dimensional evaluations from senior students (content, difficulty, workload, satisfaction) are aggregated into the AI knowledge base, offering realistic guidance during course selection.

5. Efficient administration: Online leave requests automatically sync attendance status; real-time access to schedules, grades, and personal info updates enables paperless, streamlined daily academic management.

6. Risk alerts: Intelligent attendance analysis detects prolonged absences or lateness and notifies advisors and parents proactively. Combined with credit and GPA tracking, it prevents unnoticed academic decline.

7. Data security: Students may view and edit select personal details under verified review processes. Role-based permissions strictly safeguard academic and private data.

5.3 Teachers Benefits

1. 24/7 Intelligent Q&A: LLM assistant handles over 80% of routine student and guardian inquiries on schedules, exams, and policies, freeing teachers from repetitive email responses.

2. Automated Attendance & Leave: Real-time attendance sync and smart alerts eliminate manual roll calls, leave tracking, and absence miscalculations.

3. Unified Information Portal: Single platform for class schedules, rosters, syllabi uploads, and grade management reduces system fragmentation and administrative learning curves.

4. Multi-dimensional Course Feedback: AI analyzes anonymous evaluations on teaching content, methods, difficulty, and workload to generate objective visual reports for continuous teaching improvement.

5. Smart Academic Alerts: System identifies at-risk students via performance trends and attendance, prompting timely one-on-one teacher interventions to boost class outcomes.

6. Mobile Office Flexibility: Cross-platform web and mobile access enables remote schedule viewing, administrative processing, and student communication, ideal for part-time instructors and off-campus work.

5.4 School's Management and Decision-making Level Benefits

1. Automated Processes: Full workflow automation reduces administrative workload by over 60%, minimizes manual errors, and accelerates routine academic procedures.

2. Standardized Workflows: Institutionalized, transparent operations reduce irregular practices and operational risks while strengthening overall governance capacity.

3. Quality Monitoring: Aggregated course evaluations generate multi-level visual reports, enabling management to identify issues and formulate targeted teaching improvements.

4. Data-Driven Decisions: Unified real-time data ecosystem supports analysis of enrollment, performance trends, and resource utilization for strategic planning and optimized allocation.

5. Enhanced Security: Centralized single sign-on and role-based permissions streamline access management, reduce maintenance costs, and safeguard core academic data from breaches.

6. Scalable Architecture: Microservices and standardized APIs ensure seamless future integration with existing systems and support ongoing AI capability upgrades, protecting long-term digital investment.

Section 6.0: The Project Milestones

6.1 The Alternatives of the AI-Enhanced Educational Management System

When evaluating alternative delivery strategies for the AI-Enhanced Educational Management System (AI-EMS), two principal approaches are considered: Outsourcing and In-House Development.

Outsourcing: The entire development process is contracted to an external third-party vendor. The institution defines requirements and acceptance criteria, while the vendor is responsible for design, implementation, testing, and deployment.

In-House Development: The system is built entirely by the institution's own project team, leveraging the team's technical skills and voluntary coursework efforts, with no external development contractors involved.

This subsection provides a comparative analysis of these two alternatives based on assessed value and associated risks, drawing on established software engineering and project management literature.

Value Assessment

From a value perspective, in-house development offers several advantages. First, it ensures full intellectual property ownership and absolute control over the system's technical roadmap, allowing the team to iteratively adjust features without being constrained by outsourcing contract change orders. Second, in-house development helps cultivate deep technical capabilities within the institution—the project team gains first-hand

experience in integrating large language models with academic workflows, creating long-term institutional knowledge assets. Third, for the proof-of-concept phase, the total cost of ownership for in-house development is significantly lower. Since the team's labour comes from voluntary coursework efforts, the one-time development cost is approximately USD 165, with monthly operating costs of about USD 32 (as documented in Sections 2.1 and 2.3 of this report). Outsourcing, in contrast, would require substantial vendor fees—a comparable AI-capable prototype typically carries a price tag between USD 20,000 and USD 50,000¹¹.

Outsourcing, however, may hold value in specific contexts. Professional software vendors often have mature development processes and extensive project experience, which could theoretically shorten the initial deployment timeline. For institutions lacking internal development capabilities, outsourcing shifts the management burden of recruitment, training, and staff retention to the vendor. According to Deloitte's 2024 Global Outsourcing Survey, 77% of organisations outsource IT services to some extent, with cost optimisation cited as the primary driver (68% of leaders ranked it as the top reason)¹⁶. Furthermore, some studies suggest that outsourcing can save approximately 15% on average compared to in-house operations¹¹.

Risk Comparison

The risk profiles of the two approaches differ substantially. For in-house development, the core risks are technical and managerial: AI hallucination, performance bottlenecks under high concurrency, scope creep, etc. These risks remain under the direct control of the project team and can be mitigated through rigorous design and testing.

By contrast, outsourcing introduces a range of external dependency risks that are more difficult to control. Notable risks include:

Hidden Cost Risk: The actual expenditure of an outsourced project often significantly exceeds the initial quote. Vendors may underestimate project complexity or avoid disclosing potential costs during the bidding phase. Research has shown that actual cost overruns in outsourcing projects are common, and that cost estimates tend to be systematically understated due to vendor profit maximisation¹¹. Beyond direct contract fees, outsourcing also incurs coordination costs, contract management costs, and quality control costs¹².

Vendor Lock-in Risk: Once an outsourcing model is adopted, the institution may gradually fall into technical dependency on a specific vendor. Vendors may use proprietary code frameworks, APIs, or data formats that make switching costly and technically challenging. Vendor lock-in does not necessarily start with a poor procurement decision—when a system works well, institutions can become locked in simply through increased reliance¹². Research indicates that vendor lock-in generates substantial hidden costs, including switching costs and renegotiation expenses¹³.

Data Security and Compliance Risk: Under the outsourcing model, the institution must entrust sensitive student records, teaching evaluation data, and academic information to a third-party vendor, greatly increasing the risk of data breaches and compliance violations. Earl (1996) warns against placing any information one is unwilling to lose or disclose—especially personally identifiable information (PII)—in external information systems. Outsourcing vendors may use insecure devices, have permission-revocation gaps after employee departures, or face legal compliance challenges with cross-border data transmission. Once a data security incident occurs, it not only damages the institution's reputation but may also lead to lawsuits and regulatory penalties¹⁴.

Knowledge Loss and Capability Degradation Risk: Long-term reliance on outsourced development gradually erodes the internal technical team's deep understanding and maintenance capabilities of the system. When critical system knowledge resides with the vendor rather than within the institution, any personnel changes can lead to knowledge loss and intellectual property leakage. This "capability hollowing" forces the institution to continue relying on the same vendor after the contract expires, further exacerbating vendor lock-in¹³. Scholars have noted that outsourcing also leads to a reduction in internal technical positions and negatively affects employee morale¹⁵.

Quality Control and Communication Risk. In outsourcing projects, geographical distance, cultural differences, and communication time lags between client and vendor increase the likelihood of requirement misinterpretation¹². This is particularly acute in AI education systems, where external teams struggle to quickly build sufficient domain knowledge about the institution's specific policy documents, evaluation rubrics, and workflow rules. Moreover, outsourcing contracts often fail to fully specify expected system behaviours at the edges, leading to gaps between delivered outcomes and expectations. Research has identified mismatched cost/benefit trade-offs, high hidden costs, and low responsiveness as factors contributing to outsourcing project failure¹⁵.

6.2 Recommended Solution and Reasons

Based on the comparative analysis in Section 6.1, the in-house development approach is recommended for the AI-EMS Proof of Concept (PoC). There are 5 reasons for the question that why we choose in-house development.

First, **cost efficiency**. The in-house approach requires only minimal direct monetary investment (approximately USD 165 one-time and USD 32 monthly), as the development is carried out by the student team as part of their coursework. Outsourcing would cost at least two orders of magnitude more, which is unjustifiable for an academic PoC.

Second, **full control and flexibility**. In-house development allows the team to iteratively refine features based on real-time feedback from stakeholders (administrators, teachers, and students). The agile Scrum methodology adopted by the team enables rapid adjustments to requirements, such as modifying the RAG knowledge base structure or fine-tuning the attendance geofencing logic, without contractual renegotiation.

Third, **knowledge retention and capacity building**. By building the system internally, the team acquires deep technical expertise in LLM integration, vector databases, and educational workflows. This institutional knowledge becomes a long-term asset for future AI initiatives.

Fourth, **risk controllability**. As detailed in Section 4.0, technical risks such as AI hallucination, performance bottlenecks, and scope creep can be directly managed through rigorous testing, sprint buffers, and daily stand-ups. Outsourcing would introduce external dependencies—vendor lock-in, hidden costs, data security exposure—that are harder to mitigate within the project's constrained timeline and budget.

Fifth, **alignment with project objectives**. The AI-EMS is a proof-of-concept intended to demonstrate feasibility, not a production-grade commercial system. In-house development fully satisfies the academic requirements of the course while delivering a deployable, testable prototype.

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